INCH-POUND

MIL-C-24643/4C
14 March 1994
SUPERSEDING
MIL-C-24643/4B
3 October 1986

MILITARY SPECIFICATION SHEET

CORD, ELECTRICAL, 600 VOLTS, TYPE LSMCOS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, uncoated, (see table I for size)
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Ethylene propylene rubber or cross-linked polyethylene insulation. (see table I for wall thickness).
- Sizes 2, 4 and 7 shall have special identification code applied by method 3.
- Size 5 shall have one shielded pair (colors black and white) and 3 singles (colors red, green and blue) applied by method 3.
- Size 6 shall have two shielded pairs (colors black and white for first pair and blue and red of second pair) and two singles (colors yellow and green) applied by method 3.
Fourth - Braided shield (for all pairs) of AWG no. 34 or no. 36 uncoated copper. Braid angle 30 to 35 degrees; minimum coverage of 88 percent.
Fifth - The required number of singles and shielded pairs with a left hand lay and with a minimum amount of fillers to make a round assembly. The length of lay shall be determined to permit the completed cord to withstand the twisting and bending endurance test, and for sizes 5 and 6, the mutual inductance tests. The size 7 shall have a center of glass. A single conductor shall not be used as a central core for the cord.

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Construction (non-watertight) (continued)

Sixth - In size 5 and 6 only, a binder tape applied helically with overlap in sizes 2, 4 and 7: first a binder; second, a braided shield of AWG no. 34 or AWG no. 36 uncoated strands, having a braid angle of 30 degrees minimum, and providing a coverage of 88 percent minimum; third a separator tape.

Seventh - Cross linked polyolefin jacket (black). (see table I for diameters).

* NOTE: 1. At the manufacture's option, the conductors in type and size LSMCOS-6 may consist of eight (8) strands, each having a nominal diameter of 0.010 inch, around one strand having a nominal diameter of 0.0142 inch.

* TABLE I. Details

Military part no. M24643/4	Type and size	Conductors		Insulation thickness minimum average (inch)	Overall Diameter		Voltage withstand		Insulation resistance per 1000 feet minimum (megohms)	Conductor resistance per 1000 feet maximum (ohms)
		No. prs.	Size AWG		min (inch)	max (inch)	Conductor to conductor minimum (volts)	Shield to ground minimum (volts)		
-01UO	LSMCOS-2	2	18(class K)	0.023	0.440	0.460	2000	1500	200	6.96
-02UO	LSMCOS-4	4	18(class K)	0.023	0.490	0.510	2000	1500	200	6.96
-03UN	LSMCOS-5	5	20(class K)	.013	.375	.390	1000	500	100	11.28
-04UN	LSMCOS-6	6	20(class K)	.013	.460	.480	1000	1500	100	11.28
-05UO	LSMCOS-7	7	18(class K)	.023	.575	.595	2000	1500	200	6.96

EXAMINATION AND TESTS:

Basic electrical:	Requirements:
Conductor Resistance - ohms/1000 feet at 25°C, maximum.	(see table I)
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	(see table I)
Shield to ground	(see table I)
Insulation resistance - conductor to shield, megohms/1000 feet, minimum	(see table I)
Conductor and shield continuity	No failure

Note: Shield continuity applicable to LSMCOS-2, LSMCOS-4 and LSMCOS-7 only.

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EXAMINATION AND TESTS (continued):

	Requirements
Group A:	
Visual and dimensional	No failure
* Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lb/in^2, minimum	700
Elongation - percent, minimum	250
Cross-linked polyethylene	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Twisting and bending endurance - (see note 2)	
At 75°C, cycles, minimum	6000
At minus 30 ± 2°C, cycles, minimum	2000

NOTE: 2. Roller diameters for bending endurance test shall be as follows:

Cable type and size	Roller Diameter (inch, maximum)
LSMCOS-2	1.4
LSMCOS-4	1.5
LSMCOS-5	1.2
LSMCOS-6	1.4
LSMCOS-7	1.8

Cold bending, cord at minus 40 ± 2°C ..... No failure

Group C:	
Mutual inductance (LSMCOS-5 and LSMCOS-6 only)	
Microhenries, maximum	0.0135
Physicals (aged)	
Insulation	
Ethylene propylene	
Air oven	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged minimum	75

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EXAMINATION AND TESTS (continued):

	Requirements
Cross-linked polyethylene	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cord)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
* Hot oil immersion	
Tensile strength - percent of unaged; minimum	50
Elongation - percent of unaged, minimum	50
Shrinkage	No failure
Permanence of printing (jacket) - cycles, minimum	125
Heat distortion - percent of unaged, maximum	30
Group D:	
* Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Aging and compatibility (cable) (125± 5°C.)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
* Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

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UNIT ORDERING LENGTH: 2000 feet (nominal)

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